Bashir Agro Phase 12A: API + PostGIS Backend Detailed Instructions

Phase 12A Objective

Extend the Laravel Admin Panel backend into a mobile-ready backend.
This phase focuses on Laravel REST API, authentication, database readiness and PostGIS.
Flutter UI is not included in this phase.

Scope

Included:
- REST API v1
- Laravel Sanctum
- Farmer authentication preparation
- API controllers
- API resources
- Task APIs
- Assignment APIs
- Submission backend
- PostGIS integration
- Geofence validation
- API documentation
- API testing

Not included:
- Flutter screens
- Mobile offline storage
- Firebase mobile integration

Architecture

Admin Panel
|
Laravel Backend
|
REST API
|
Flutter Farmer App (future)
|
PostgreSQL + PostGIS

API Structure

Prepare:
routes/api.php

Version:
/api/v1/

Controllers:
- AuthController
- FarmerController
- TaskController
- AssignmentController
- SubmissionController
- LocationController

Authentication

Use Laravel Sanctum.

Flow:

Farmer App
|
Login API
|
Laravel validation
|
Token generation
|
Authenticated requests

Database Preparation

Prepare relationships:

User
|
Farmer Profile

Assignments connect:
Farmer + Task Template + Location

Submission connects:
Assignment + Farmer + Evidence

Submission System

Create task_submissions table.

Fields:
- assignment_id

- farmer_id
- notes
- photos
- latitude
- longitude
- accuracy
- status
- submitted_at
- verification_status

PostGIS and Geofence

Enable PostgreSQL PostGIS.

Store:
- Farm coordinates
- Spatial geometry

Validation:

Farmer GPS
|
Distance calculation
|
Compare radius
|
Accept or reject submission

Testing

Test:
- Authentication API
- Task retrieval
- Assignment retrieval
- Submission API
- Geofence validation

Antigravity Instruction

Implement Phase 12A only.

Do not build Flutter.

Implement Laravel API, Sanctum, PostGIS, submission backend, documentation and tests.

Preserve existing Admin Panel functionality.

Run:

php artisan test
vendor/bin/pint

Report files, database changes, APIs and test results.